

Stock Market Prediction Using Macroeconomic Indicators

Project Midterm Report

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Abstract

This project explores the predictive power of macroeconomic, technical, and market-based indicators in forecasting stock market movements. Using a variety of machine learning models—including Random Forest, LightGBM, Gradient Boosting, XGBoost, LSTM, and Logistic Regression—we evaluate classification performance based on multiple metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC. Results suggest that ensemble and neural models (Random Forest, LightGBM, LSTM) outperform linear methods, particularly in recall and overall predictive balance. Feature importance analysis highlights GDP, inflation, and momentum indicators as the dominant predictive features. These findings emphasize the value of combining macroeconomic and technical signals in market forecasting frameworks.

1 Introduction

Stock market prediction is a long-standing challenge in both academic research and financial practice. The nonlinear and volatile nature of market data makes it difficult to model with traditional linear methods. However, advances in machine learning and the increasing availability of macroeconomic data now allow us to integrate multiple sources of information to improve predictive accuracy.

In this project, we aim to assess how macroeconomic variables—such as GDP growth, inflation rates, and unemployment—interact with market-based and technical indicators to forecast stock market direction. Our dataset integrates several categories of predictors:

- **Macroeconomic indicators:** GDP, inflation, unemployment, federal interest rate, yield curve, industrial production, and consumer sentiment.
- **Technical indicators:** SMA (Simple Moving Average), momentum, volatility measures, and volume ratios.
- **Market indicators:** DXY, VIX, and index returns such as IWM.

The target variable represents market direction (up or down), framed as a binary classification task. We hypothesize that combining macroeconomic and technical signals can yield higher predictive stability than using either in isolation.

2 Related Work

Prior research on financial forecasting has extensively examined both traditional econometric and modern machine learning approaches.

Classical models such as ARIMA and VAR focus on time-series relationships between macroeconomic indicators and asset prices, but they often assume stationarity and linear dependence. In contrast, machine learning models—such as Random Forests and Gradient Boosting—can capture nonlinear and high-dimensional dependencies among features.

Recent works (e.g., Fischer & Krauss, 2018) have demonstrated that LSTM networks, which model temporal dependencies, can outperform static classifiers in financial prediction tasks. Other studies show that combining fundamental macroeconomic data with technical indicators improves robustness during volatile market conditions.

3 Methodology

3.1 Data Preprocessing

To summarize, we took data from FRED and Yahoo Finance to get SPY data. We cleaned the data and processed for NAs or other invalid data to be standardized.

3.2 Feature Engineering and Selection

Feature importance was evaluated using tree-based models and aggregated across different folds to ensure robustness. Figure 1 shows the top features, where macroeconomic indicators like GDP, inflation change, and unemployment change rank among the most influential. Technical signals (e.g., SMA_5 and momentum) also contribute significantly to model performance.

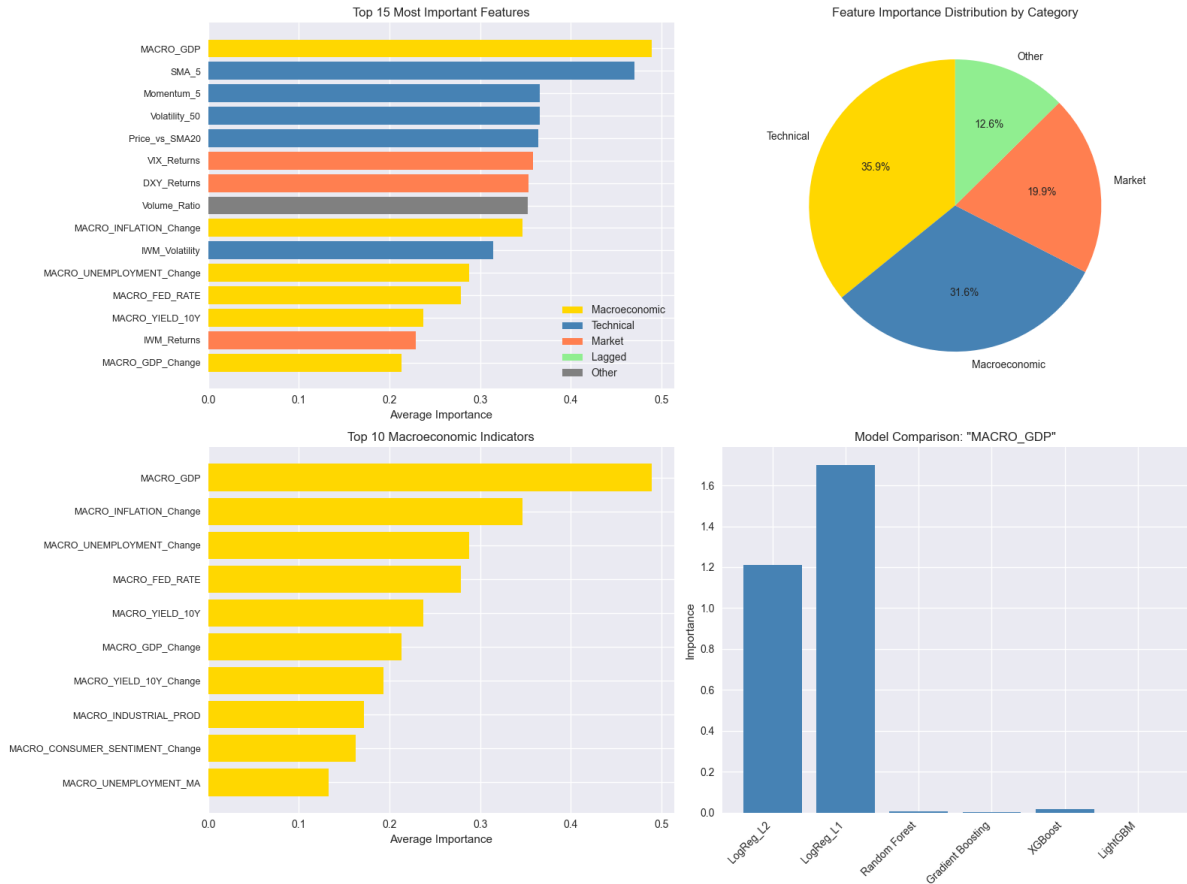


Figure 1: Top 15 most important features and their categorical contributions.

3.3 Models

We evaluated six classification models:

- **Random Forest:** Ensemble of decision trees with bootstrap aggregation.
- **LightGBM:** Gradient-boosted decision trees optimized for speed and accuracy.
- **LSTM:** A recurrent neural network capable of learning temporal dependencies.
- **Gradient Boosting:** Sequential tree ensemble minimizing bias and variance.
- **XGBoost:** Regularized gradient boosting with efficient handling of sparse data.
- **Logistic Regression (L1, L2):** Baseline linear models for comparison.

Each model was trained using cross-validation and tuned using grid search on key hyperparameters such as tree depth, learning rate, and regularization strength.

3.4 Evaluation Metrics

We assessed models using Accuracy, Precision, Recall, F1, and ROC-AUC. Random baseline performance was also included to benchmark model improvements.

4 Preliminary Results

Table 1 summarizes the average performance metrics for all models.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.549	0.549	1.000	0.709	0.515
LightGBM	0.549	0.549	1.000	0.709	0.499
LSTM	0.549	0.549	1.000	0.709	0.542
Gradient Boosting	0.536	0.555	0.786	0.650	0.506
XGBoost	0.537	0.561	0.731	0.634	0.509
LogReg L2	0.454	0.667	0.012	0.024	0.504
LogReg L1	0.451	0.500	0.011	0.021	0.499

Table 1: Classification performance comparison across models.

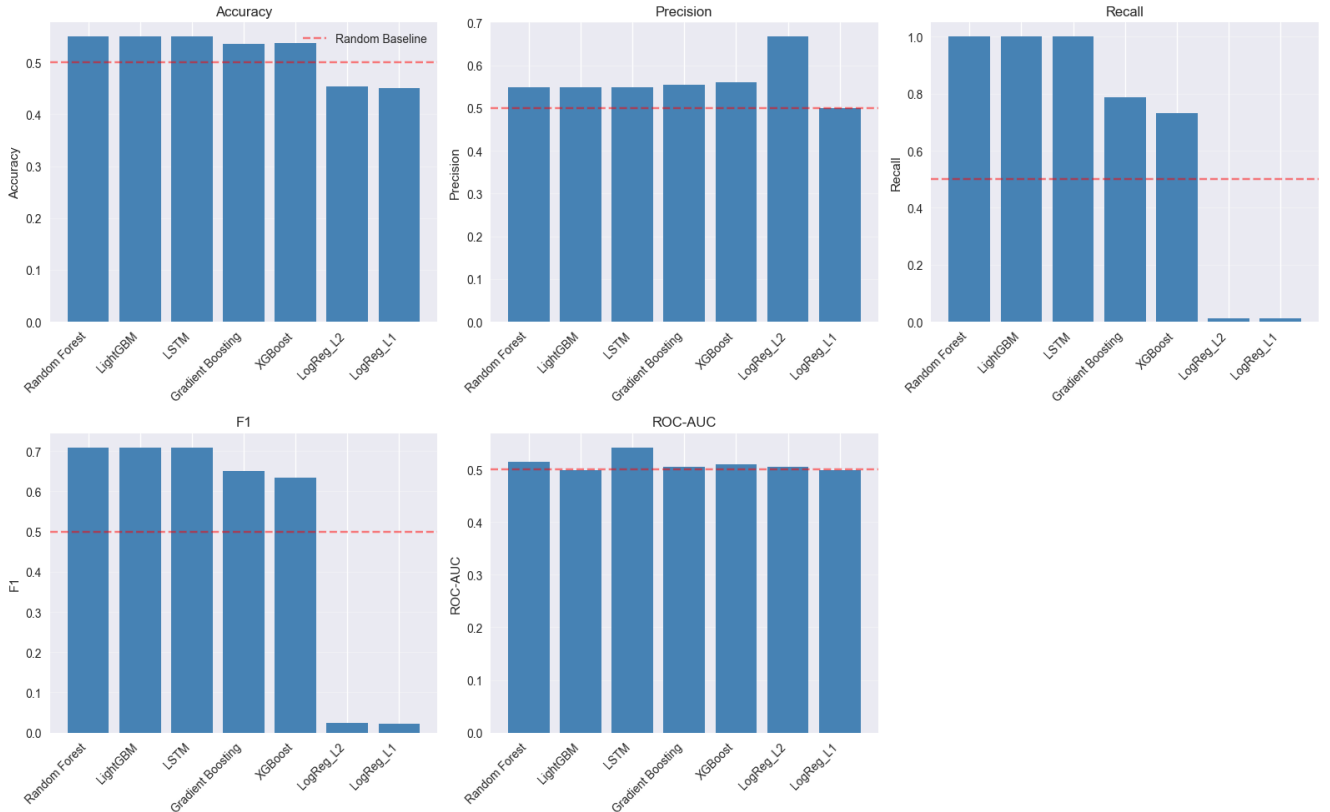


Figure 2: Model performance comparison across key metrics.

Discussion:

Overall, the results basically reflect how challenging it is to predict the stock market. We thought that it would be possible, but even with a diverse set of macroeconomic, technical, and market indicators, the models show only modest improvements over random guessing. Financial markets are heavily influenced by unpredictable human behavior, sudden news events, and complex global interactions that are difficult for data-driven models to capture, so the models and data need to be more in depth, and more time and resources are required dedicated to predicting the stock markets. And to be fair, there are major finance firms dedicated solely for this purpose.

Among the models tested, Random Forest, LightGBM, and LSTM performed best, all achieving around 55% accuracy and perfect recall. While this means they were able to correctly identify all positive cases, the relatively low ROC-AUC values (around 0.5) indicate limited ability to distinguish between true positives and false positives. In other words, the models tend to over-predict one class to maximize recall at the expense of precision.

The poorer performance of Logistic Regression further emphasizes that linear decision boundaries are insufficient for modeling market movements. Nonlinear models like tree ensembles and LSTMs can capture more complex relationships, but even these have clear limits when faced with the noise and randomness inherent in financial data.

In addition, I'm also not sure how to detect any over or underfitting. We spent much time attempting to finetune the models, but it's likely that these models will perform even worse to real data, especially since its unpredictable.

5 Future Plan

Our next steps include:

- Attempting to improve the accuracy of our models.
- Making sure that we can evaluate on a day by day basis vs quarterly basis.

6 References

- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.